Multicollinearity: Detection and Remedies

Applied Linear Models I • Lecture 10 of 16

Today

- **Detect**: VIF, condition number, eigenstructure.
- **Consequence**: inflated variance, unstable coefficients (not bias).
- **Ridge** regression – smooth shrinkage.
- **PCR** – truncation to high-variance directions.
- A **bias–variance trade-off** that can beat OLS on prediction.

Detection (longley)

- **VIF**$_j = 1/(1 - R_j^2)$: GNP's VIF > 1000 (variance $\times 1000$).
- **Condition number** ≈ 81 (> 30 is serious).
- **Eigenvalues** of scaled $\mathrm{X}^\top \mathrm{X}$: 3.63, 1.17, 0.20, 0.005, 0.0006.

Tiny eigenvalues = near-degenerate directions: combinations of predictors that barely vary and are poorly estimated.

Consequence: instability (not bias)

OLS stays **unbiased** and predicts well in-sample. What breaks is **precision and stability**:

Tiny perturbation, big swing

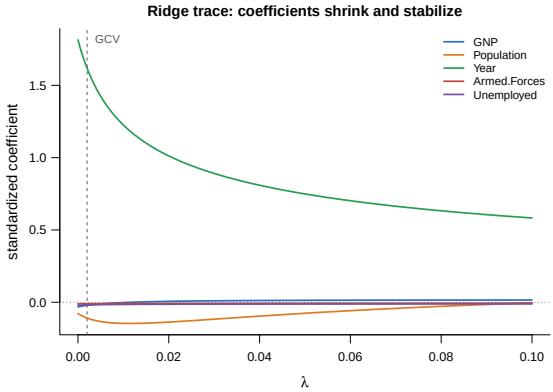
Add noise (SD 0.1) to `Employed`: the GNP coefficient swings $\approx 25\%$. Near-singular $X^\top X$ amplifies any wiggle.

This is why individual t's can be insignificant while the overall F is highly significant.

Ridge regression

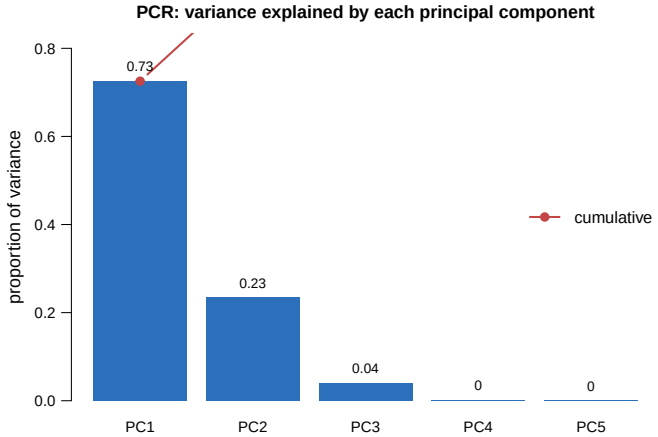
$$\hat{\boldsymbol{\beta}}_\lambda = \arg\min_{\boldsymbol{\beta}} \ \|\mathrm{y} - \mathrm{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|^2 = (\mathrm{X}^\top\mathrm{X} + \lambda\mathrm{I})^{-1}\mathrm{X}^\top\mathrm{y}.$$

- Lifts tiny eigenvalues off zero.
- **Biased**, lower variance.
- ℓ_2 constraint $\|\boldsymbol{\beta}\| \le t$.
- Choose λ by GCV / CV.



Principal-components regression

Regress y on the first m principal components; drop low-variance directions.



Ridge vs PCR; bias–variance

- PCR **truncates** (weight 1 or 0); ridge **shrinks** every direction, hardest on low-variance ones. Ridge = smooth PCR.

Ridge can beat OLS out of sample

Two collinear predictors: test MSE OLS = 1.24 vs ridge = 0.20 – a sixfold gain. Accept bias, cut variance.

What they do NOT fix

They stabilize estimation; they cannot create missing information. If two predictors measure the same thing, combine, drop, or collect more data.

Summary

- Detect with VIF, condition number, eigenvalues.
- Collinearity inflates variance and destabilizes coefficients – not bias.
- Ridge: smooth shrinkage; PCR: truncation; tune by CV.
- Both trade bias for variance and can win on prediction MSE.

Next

Lecture 11: qualitative predictors and ANCOVA.